MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 10: Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If angle between two radii of a circle is 130°, then the angle between the tangents at the ends of these radii is: [1]

(A) 90° (B) 50° (C) 70° (D) 40°

Q2. The pair of tangents AP and AQ drawn from an external point A to a circle with centre O are perpendicular to each other and length of each tangent is 5 cm. The radius of the circle is: [1]

(A) 10 cm (B) 7.5 cm (C) 5 cm (D) 2.5 cm

Q3. Two concentric circles have radii 13 cm and 5 cm. The length of the chord of the larger circle which touches the smaller circle is: [1]

(A) 12 cm (B) 24 cm (C) 18 cm (D) 20 cm

Q4. If tangents PA and PB from a point P to a circle with centre O are inclined at 60° and $OP = 10$ cm, then length of tangent PA is: [1]

(A) 5 cm (B) $5\sqrt{3}$ cm (C) $10\sqrt{3}$ cm (D) $5/\sqrt{3}$ cm

Q5. **Assertion (A):** A parallelogram circumscribing a circle is always a rhombus.
Reason (R): In any quadrilateral circumscribing a circle, the sum of opposite sides is equal, so $AB + CD = AD + BC$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre. [3]

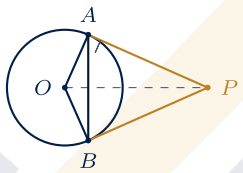
Q8. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at point T. Find the length of tangent TP. [3]

Q9. A quadrilateral $ABCD$ circumscribes a circle with centre O. Prove that opposite sides subtend supplementary angles at the centre, i.e., $\angle AOB + \angle COD = 180°$. [3]

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Highway Traffic Roundabout Slip-Lane Design

Civil engineers design a circular traffic roundabout of radius $r = 15$ m centred at O. Two straight slip lanes PA and PB meet at an intersection point P outside the roundabout and touch the circular island at points A and B. The angle between the slip lanes is $\angle APB = 60°$.



(i) Find the measure of the central angle $\angle AOB$ subtended by the slip-lane entry points. [1]

(ii) Calculate the straight-line distance OP from the intersection to the roundabout centre. [1]

(iii) Calculate the length of each slip lane PA and PB, OR prove that $\triangle PAB$ forms an equilateral triangle. [2]

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